

Diabetic Retinopathy Classification Using Deep Learning Techniques



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1. Systematic Literature Review

1.1. Databases Used

Databases	Links
IEEE Xplore	https://ieeexplore.ieee.org/
PubMed	https://pubmed.ncbi.nlm.nih.gov/
SpringerLink	https://link.springer.com/
ScienceDirect	https://www.sciencedirect.com/
Google Scholar	https://scholar.google.com/

1.2. Search strings

These search strings were used to retrieve relevant studies from selected databases:

Search string ID	Search strings
1	("diabetic retinopathy classification" AND "transfer learning" OR "CNN")
2	("diabetic retinopathy" AND "hybrid deep learning")
3	("diabetic retinopathy" AND "attention-based hybrid deep learning")

Table 1: Search Strings

1.3. Retrieved Articles from Databases

A systematic search was performed using predefined search strings defined in Table 1 for the retrieval of articles from multiple databases, including IEEE Xplore, PubMed, SpringerLink, ScienceDirect and Google Scholar. The search retrieved a total of 308 articles as shown in the figure 1. ScienceDirect has the highest number of records (95), followed by Google Scholar (90). SpringerNature retrieved 69 records, while IEEE Xplore yielded 46

records. PubMed with the fewest articles, retrieved only 8 records. Overall, this initial search yielded a significant number of studies related to diabetic retinopathy classification using deep learning and hybrid techniques, which were then considered for duplicate removal, screening and final study selection as part of the systematic review process.

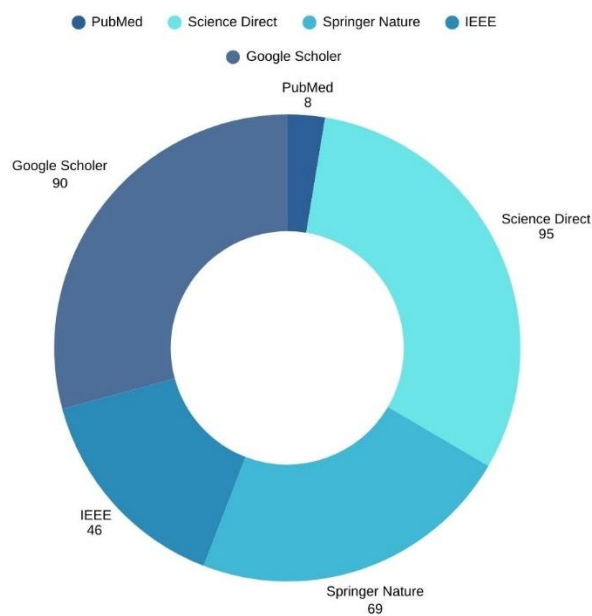


Figure 1: Number of Retrieved Articles from Databases

1.4. Inclusion and Exclusion Criteria

1.4.1. Inclusion Criteria

- Studies mainly focusing on Multi Class diabetic retinopathy (DR) classification.

- Studies implementing hybrid deep learning techniques (e.g., CNN with attention, feature fusion, transformers, or vision transformer).
- Research mostly done on well-known fundus images datasets.
- Journal articles, conference papers, or high-impact open access publications.
- Articles published between 2023–2025.
- Studies written in the English language.

1.4.2. Exclusion Criteria

- Studies not related to diabetic retinopathy classification.
- Research using only traditional machine learning techniques without hybrid or deep learning models.
- Reports, review articles, theses, posters
- Studies that do not use publicly available datasets.
- Articles published before 2023.
- Studies not written in the English language.

1.5. Selection of Relevant Studies

This review follows the PRISMA guidelines for searching and selection of articles[\[1\]](#) . We searched for 308 related papers published from 2023 to 2025 through IEEE Xplore, PubMed, SpringerLink, ScienceDirect and Google Scholar.

After removing 30 duplicate records using the reference management software Mendeley, 278 unique studies were kept for screening. During the title and abstract screening phase, 200 articles were excluded for not meeting the inclusion criteria. This screening

narrowed the selection to 78 full-text articles for detailed evaluation. Following the full-text review, 40 articles were excluded due to unsuitable datasets, absence of hybrid deep learning techniques or lack of focus on DR classification. After full text review, 38 articles were retained but 8 were excluded after further evaluation. Finally, 30 high-quality studies were selected for inclusion in this review based on their strong alignment with the research objectives. The flow diagram of the articles searching and selection is shown in Figure 2.

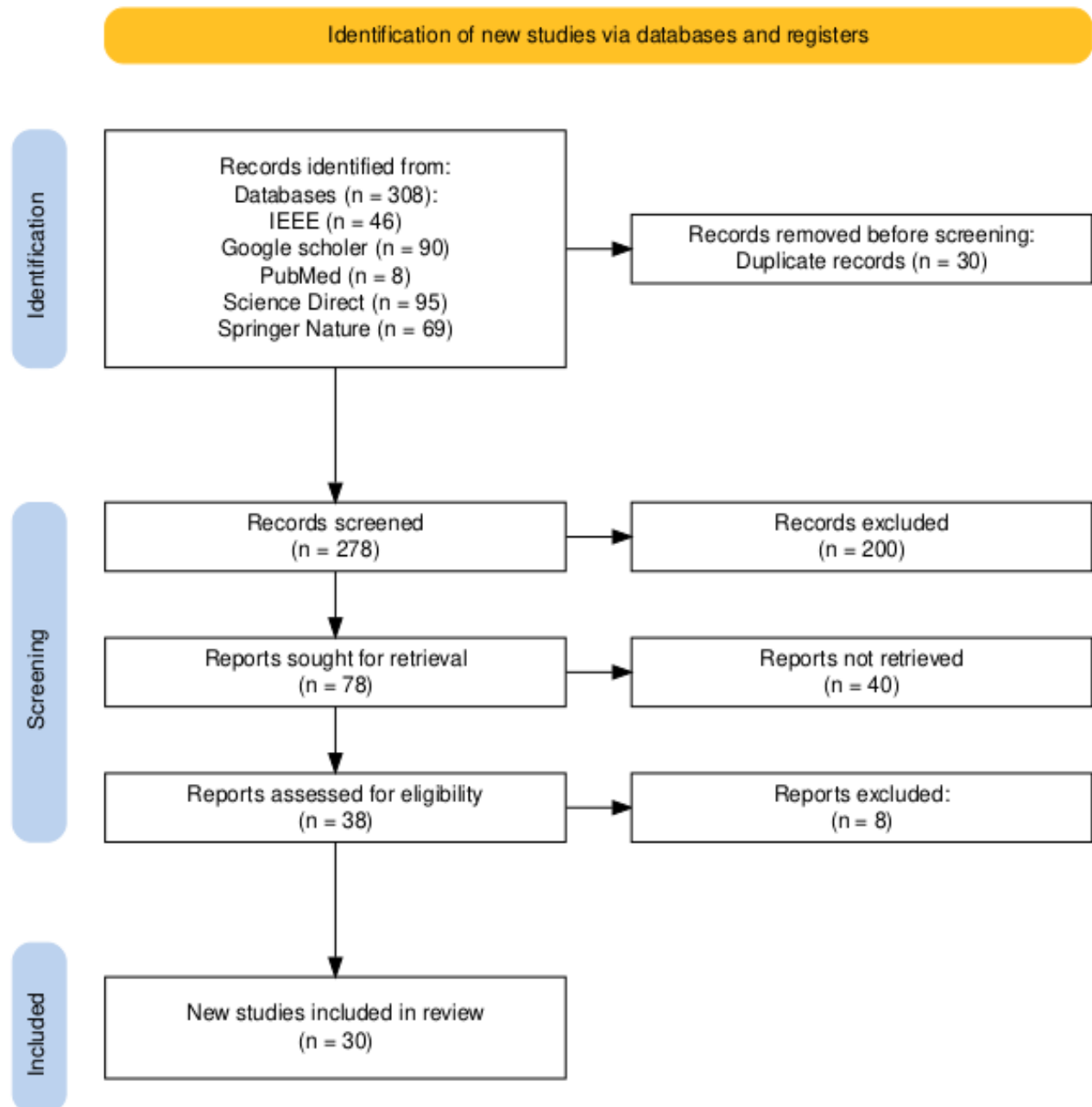


Figure 2: Prisma Flow Diagram

- [1] "PRISMA 2020 statement — PRISMA statement." Accessed: Dec. 14, 2025. [Online]. Available: <https://www.prisma-statement.org/prisma-2020>